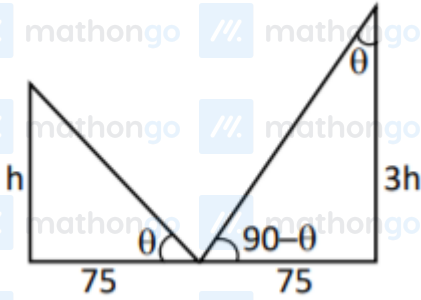


Q1 (4)

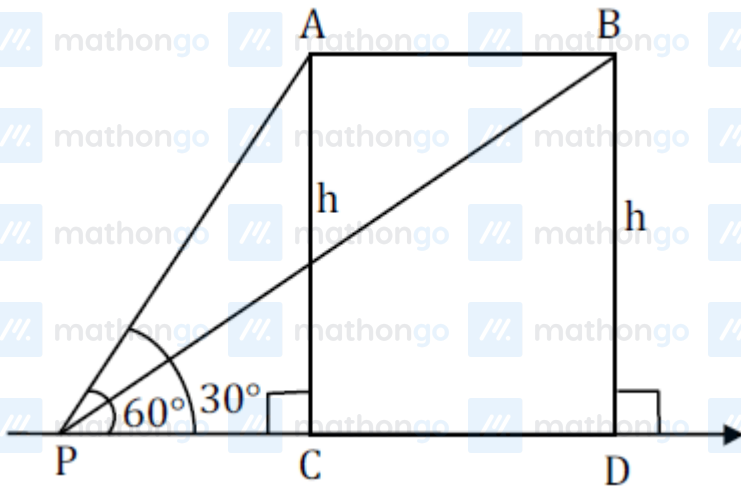
$$\tan \theta = \frac{h}{75} = \frac{75}{3h}$$

$$\Rightarrow h^2 = \frac{(75)^2}{3}$$

$$h = 25\sqrt{3} \text{ m}$$



Q2 (1)



$$v = 432 \times \frac{1000}{60 \times 60} \text{ m/sec} = 120 \text{ m/sec}$$

$$\text{Distance } AB = v \times 20 = 2400 \text{ meter}$$

In $\triangle PAC$

$$\tan 60^\circ = \frac{h}{PC} \Rightarrow PC = \frac{h}{\sqrt{3}}$$

In $\triangle PBD$

$$\tan 30^\circ = \frac{h}{PD} \Rightarrow PD = \sqrt{3} h$$

$$PD = PC + CD$$

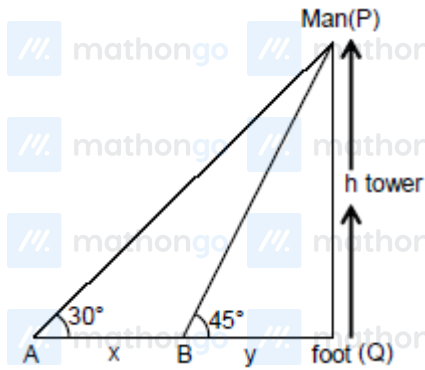
Hints and Solutions

MathonGo

$$\sqrt{3}h = \frac{h}{\sqrt{3}} + 2400 \Rightarrow \frac{2h}{\sqrt{3}} = 2400$$

$$h = 1200\sqrt{3} \text{ meter}$$

Q3 (4)



$$\frac{h}{x+y} = \tan 30^\circ$$

$$x + y = \sqrt{3}h$$

Also

$$\frac{h}{y} = \tan 45^\circ$$

$$h = y$$

$$\text{put in (1)} \quad x + y = \sqrt{3}y$$

$$x = (\sqrt{3} - 1)y$$

$$\frac{x}{20} = 'v' \text{ speed}$$

$\therefore$ time taken to reach

Foot from B

$$\Rightarrow \frac{y}{v}$$

$$\Rightarrow \frac{x}{(\sqrt{3}-1) \cdot x} \times 20$$

$$\Rightarrow 10(\sqrt{3} + 1)$$

Q1: 24 Feb (Shift 1) - Single Correct

Two vertical poles are 150 m apart and the height of one is three times that of the other. If from the middle point of the line joining their feet, an observer finds the angles of elevation of their tops to be complementary, then the height of the shorter pole (in meters) is:

- (1) 25
- (2) $20\sqrt{3}$
- (3) 30
- (4) $25\sqrt{3}$

Q2: 24 Feb (Shift 2) - Single Correct

The angle of elevation of a jet plane from a point A on the ground is 60° . After a flight of 20 seconds at the speed of 432 km/ hour, the angle of elevation changes to 30° . If the jet plane is flying at a constant height, then its height is:

- (1) $1200\sqrt{3}$ m
- (2) $1800\sqrt{3}$ m
- (3) $3600\sqrt{3}$ m
- (4) $2400\sqrt{3}$ m

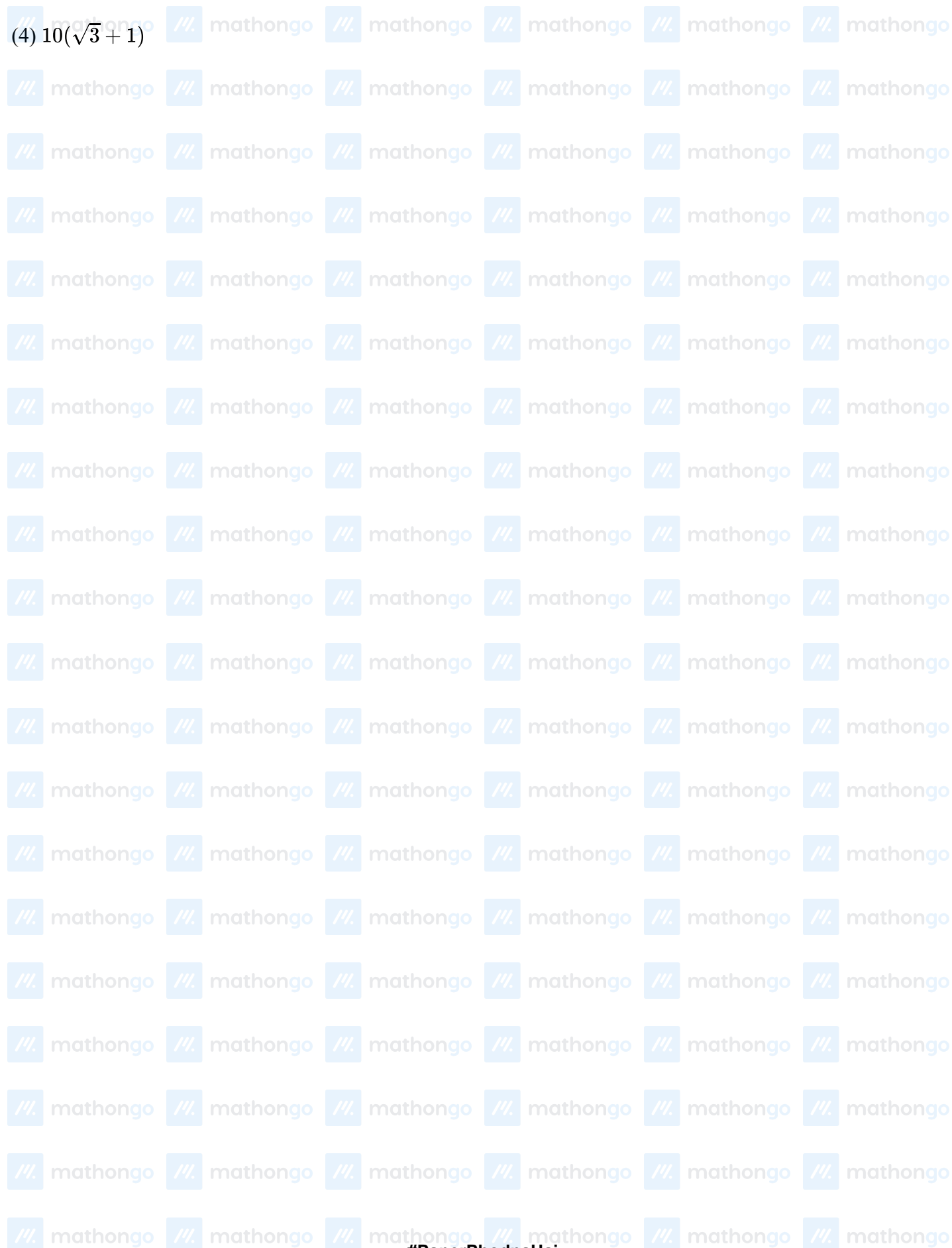
Q3: 25 Feb (Shift 1) - Single Correct

A man is observing, from the top of a tower, a boat speeding towards the tower from a certain point A, with uniform speed. At that point, angle of depression of the boat with the man's eye is 30° (Ignore man's height).

After sailing for 20 seconds towards the base of the tower (which is at the level of water), the boat has reached a point B, where the angle of depression is 45° . Then the time taken (in seconds) by the boat from B to reach the base of the tower is :

- (1) $10(\sqrt{3} - 1)$
- (2) $10\sqrt{3}$
- (3) 10

(4) $10(\sqrt{3} + 1)$



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Answer Key

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Q1 (4)

Q2 (1)

Q3 (4)

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